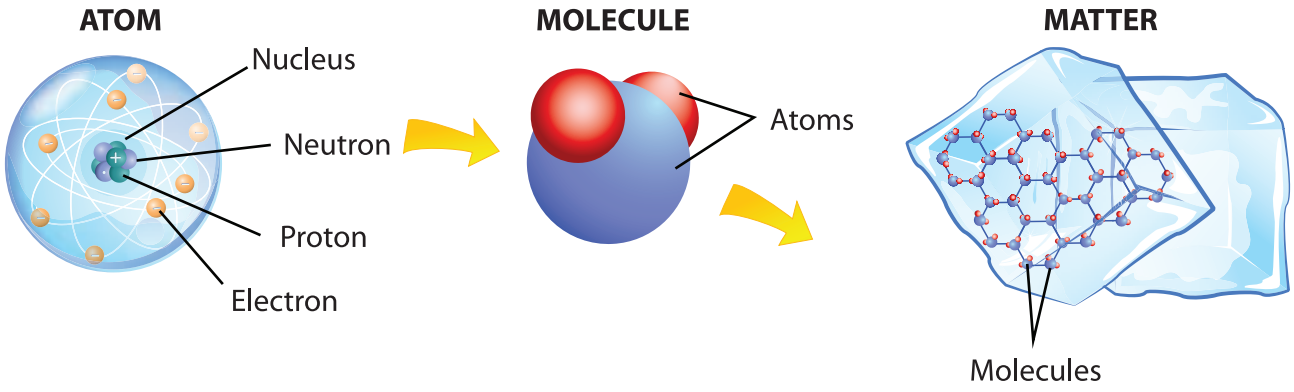


Characteristics of the particles of matter

1. Particles of matter have a lot of space in between them. In different forms of matter this spacing will be different.

Let us add a spoon full of sugar to a glass of water. Stir well. Sugar disappears completely. Where has it gone? Will the glass of water be now sweet? Water particles have space between them and sugar particles are now occupying those spaces.



2. Particles of matter attract each other.

It is the force of attraction which keeps the particles together. This attractive force will be different for different forms of matter.

Grouping of Matter on the basis of Physical states

These are the three physical states of matter. Matter can be grouped into solids, liquids and gases based on the above characteristics.

3.2 Mass, Shape and Volume of Solids, Liquids and Gases

Let us first take any solid say a stone: Answer the following questions.

❖ Do you need a container to know the shape of a stone ? Yes / No

A solid does not need a container. It stays as it is because its particles are tightly packed and has a definite shape.

❖ If you move the stone from the ground to a table or place it on the shelf does it's shape change? Yes / No

If you take a stone from the ground and place it on the table or shelf its shape and volume do not change.

Activity 2

Sit together in groups of three. Look at the objects given below. Are they familiar to you? Are they same or different? On what basis you can group them? Is there only one way of doing it or more ways? Discuss with your group members and note down your points.

Pencil and books are used for studying. The bucket and the comb are made of plastic while the table and ladle